The empirical recurrence for OEIS A250821, proved by transfer matrix

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3 September 2026

Abstract

OEIS A250821 counts arrays over $\{0,1\}$ in which a statistic taken over short runs of cells produces a derived array that is monotone in a named direction. The entry carries an empirical recurrence of order 29 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical: every comparison the condition makes lies inside a bounded window of consecutive rows, so the count is a walk count on $S = 2744$ states, and the conjectured recurrence is then settled by a finite exact computation.

2020 Mathematics Subject Classification. 05A15, 15A18.

1 The conjecture

OEIS A250821 is "Number of (n+2)X(2+2) 0..1 arrays with nondecreasing maximum minus minimum of every three consecutive values in every row and column.". It has offset 1 and begins

$$2744, 22874, 176184, 1217728, 8095062, 52143132, 330659096, 2074791394, \ \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = 9*a(n-1) -5*a(n-2) -91*a(n-3) +5*a(n-4) +431*a(n-5) +799*a(n-6) -1199*a(n-7) -4097*a(n-8) -1157*a(n-9) +8766*a(n-10) +11816*a(n-11) -4685*a(n-12) -20537*a(n-13) -12267*a(n-14) +11123*a(n-15) +19714*a(n-16) +5392*a(n-17) -8559*a(n-18) -8485*a(n-19) -695*a(n-20) +2925*a(n-21) +1310*a(n-22) -248*a(n-23) -338*a(n-24) +36*a(n-25) +56*a(n-26) -16*a(n-27) -3*a(n-28) +a(n-29)

As of the "Last modified" line on the live entry (Jul 23 2025 12:46:46, revision 6) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

The array has 4 cells across and entries $x(i,j) \in \{0,1\}$.

Fix the statistic $\sigma(v) = \max_t v_t - \min_t v_t$ of a run v of 3 consecutive cells — in words, the maximum minus the minimum of the 3 values. Two derived arrays are formed and both must be nondecreasing in their own direction: reading along a row,

$$\sigma\big(x(i,j),\ldots,x(i,j{+}2)\big) \ \le\ \sigma\big(x(i,j{+}1),\ldots,x(i,j{+}3)\big),$$

and reading down a column,

$$\sigma\big(x(i,j),\ldots,x(i{+}2,j)\big) \ \le\ \sigma\big(x(i{+}1,j),\ldots,x(i{+}3,j)\big).$$

The same statistic and the same run length serve both directions, so the condition is unchanged by transposing the array.

Nothing in the condition is invariant under renaming the letters — each statistic either does arithmetic on them or compares them by size — so no reduction to equality patterns is available, and the walk is run over the alphabet the entry names.

3 A bounded window

Lemma 1. *The row comparisons involve one array row each. A column comparison involves the* $3+1=4$ *consecutive rows* $i,\dots,i{+}3$ *and no others.*

Proof. The two runs compared start at rows i and $i+1$ and have length 3, so together they occupy rows i through $i+3$. $\square$

So the window of the last 3 rows is a state, each row of it already satisfying its own row comparisons, and one step appends a row and settles the column comparisons that the new row completes. With M the adjacency matrix of that digraph and $\iota=\tau=\mathbf{1}$,

$$a(n) \;=\; \iota^{\top} M^{\,j}\tau, \qquad S=2744,$$

a walk of no steps being an array of exactly 3 rows.

The walk index j and the entry's own n differ by a constant, read off by matching the published terms rather than assumed. In particular a is C-finite.

4 The proof

Lemma 2. *Let* $q(t)=t^{29}-\sum_i c_i t^{29-i}$ *be the characteristic polynomial of the conjectured recurrence. The residual* $a(n)-\sum_i c_i a(n-i)$ *equals* $\iota^{\top}M^{\,j}q(M)\tau$ *for the corresponding walk index* j*, so the recurrence holds from some point on if and only if* $\iota^{\top}M^{j}q(M)\tau=0$ *for all large* j*, and it is enough to examine* $j<S$*.*

Proof. Factor M^j out of $M^{j+29}-\sum_i c_i M^{j+29-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j=\iota^{\top}M^{j}q(M)\tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. $\square$

Theorem 3.

$$a(n) \;=\; 9\,a(n{-}1){-}5\,a(n{-}2){-}91\,a(n{-}3){+}5\,a(n{-}4){+}431\,a(n{-}5){+}799\,a(n{-}6){-}1199\,a(n{-}7){-}4097\,a(n{-}8){-}1157$$

for every $n>29$.

Proof. The vector $w=q(M)\tau$ was formed by 29 matrix–vector products in exact integer arithmetic and $\iota^{\top}M^{j}w$ evaluated for $j=0,1,\dots,$ again exactly; those integers vanish from the index corresponding to $n=29+1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. $\square$

5 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 17 terms the entry publishes, exactly, in integer arithmetic.

Second, the count was recomputed for the smallest arrays straight from the definition: fill the cells in row major order and test each comparison the moment both of its runs are complete, in the orientation the entry's own name writes. No transfer matrix and no transposition are involved, and the published terms came back.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 2 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A250821.

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[3] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.